

2026-06-10-llms-access-master-roadmap

Master Roadmap — Models-Access Refinement & Agents-Work Enforcement

Field	Value
Revision	4
Last modified	2026-06-10
Status summary	DELIVERED — SP0–SP7 implemented across 9 waves (~33 commits, all RED→GREEN + §1.1 + reviewed + pushed). Model-access end-to-end; unified /v1/catalog with real verified models; full per-agent CLI-bridge de-bluff (68 pkgs); 49 forks + swap + GitLab mirrors; SDK currency incl. security; substrate/D-7; 4 local harnesses; CONST-066 doc compliance; G-2 repos. Defects D-1..D-20 (18 fixed). Remaining: tree-sitter migration (in flight, feasibility-first) + boot-infra (§12.6 host-safety-gated). See docs/CONTINUATION.md close-out ^{162/163} for the full resumption record.
Source request	docs/requests/helix_code_request_llms_access.txt
Delivery model	Independent sub-programs (operator decision 2026-06-10) — this doc is the index/roadmap; each sub-program gets its own spec → plan → release cycle
Authority	Cascades from constitution/ + root CLAUDE.md/CONSTITUTION.md; anti-bluff §11.4 family governs every closure

Anti-bluff note (§11.4 / §11.4.123): This roadmap is a *plan*, not a claim of done work. No phase below is “complete” until it carries captured runtime evidence per §11.4.5 / §11.4.69 / §11.4.107. The brainstorming HARD-GATE applies: **no implementation, fork creation, or push happens until each sub-program’s design+plan is approved.**

0. Operator decisions baked in (2026-06-10)

- Fork strategy:** Fork ALL ~50 cli_agents/ * third-party submodules **now** under the vasic-digital org with prefix caf- (cli-agent-fork) → e.g. git@github.com:vasic-digital/caf-claude-code.git. Swap each submodule’s URL to our fork. Auto-merge-from-upstream for every fork.
- Delivery:** Independent sub-programs; this is the master index.

3. **Bridge exposure: One provider per CLI agent** — each working CLI agent surfaces its own dynamically-discovered model list + power-features (Vision / Generative / RAG / Memory / MCP / LSP / ACP).
 4. **New “ways-of-working” submodules** (parallel-agents, dynamic-flows, ...): under the **HelixDevelopment** org.
 5. **Constitution DESCOPED (2026-06-10)**: no governance-rule authoring, no constitution push — **sync only** (fetch+pull to current). Reusable fork/merge scripts live **local to HelixCode** (scripts/caf/), not in constitution/scripts/.
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1. Architecture grounding (verified from repo, 2026-06-10)

- **HelixAgent** is the “root” the request extends — submodules/helix_agent/internal/ already contains ensemble, llm, router, verifier, providers, modelsdev, models, clis, mcp, rag, memory, agentic, agents, concurrency, planning, background. (Verified by ls.)
- **LLMsVerifier** (submodules/llms_verifier) is the single source of truth for working models (CONST-036/040) — all dynamic discovery flows through it; hardcoded model lists are violations.
- **Inner Go app** (helix_code/) has internal/llm/*_provider.go (anthropic, azure, bedrock, copilot, deepseek, gemini, groq, ...), internal/provider, internal/providers, internal/agent.
- **Constitution scripts** live in constitution/scripts/ (codegraph, action-prefix, subagent-tiering, token-accounting, workable-items, hooks). **No fork-maintenance scripts exist yet** — Sub-program SP0 adds them.
- **~50 CLI agents** under cli_agents/ as third-party submodules (verified .gitmodules).

Per-workstream evidence-backed analyses (produced by 6 parallel subagents, 2026-06-10) land in docs/superpowers/specs/analysis/2026-06-10-{A..F}-*.md and feed the detailed sub-program specs.

1a. Real defects surfaced during planning (evidence-backed — to be fixed inside the relevant SP, RED-first §11.4.115)

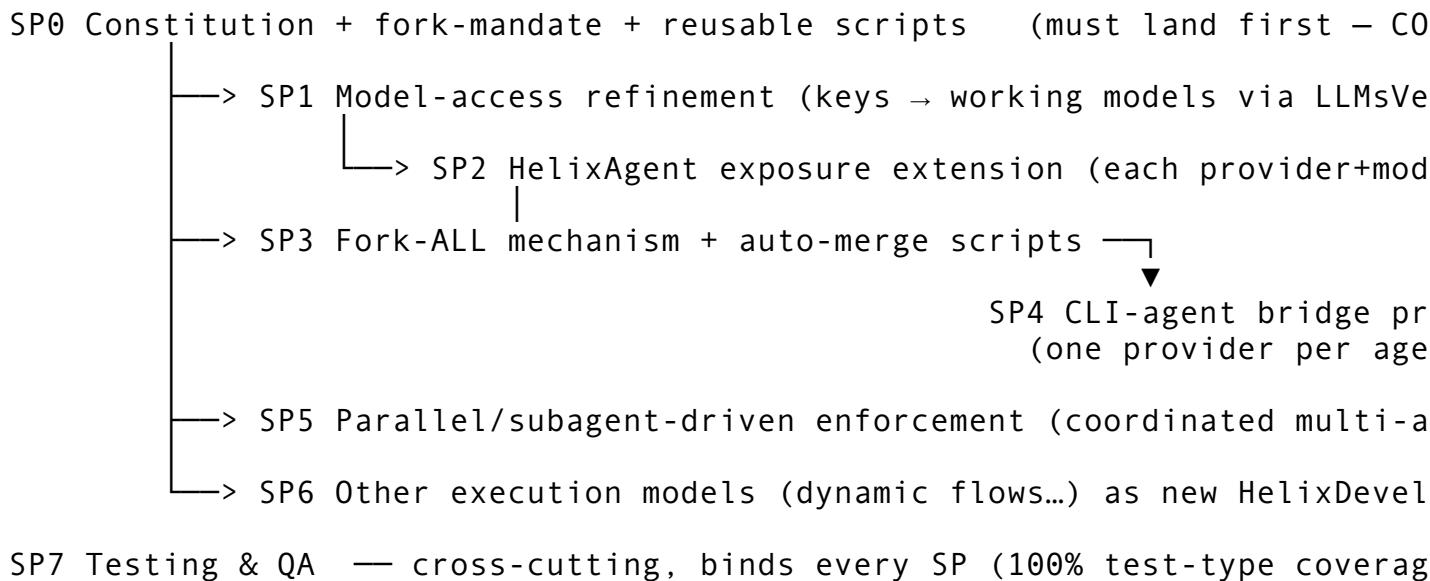
These are concrete §11.4 anti-bluff / CONST violations found by the analysis subagents with file:line evidence — independent of the new features, they are pre-existing defects:

ID	Defect	Evidence	Owner SP
D-1	/v1/completion/models returns a hardcoded 3-model list	helix_agent/internal/handlers/completion.go:406	SP2 (CONST-036/ BLUFF-002)
D-2	CLI handleListModel lists failed/pending models as available	helix_code/.../cmd/cli/main.go:1361	SP1
D-3	secrets.LoadAPIKeys is DEAD code (never called in prod)	helix_code/internal/secrets/loader.go:30	SP1

ID	Defect	Evidence	Owner SP
D-4	“Working-model” filter (Verified $\wedge$ score $\geq$ min) loaded but never applied	adapter.go:175	SP1
D-5	Hardcoded provider model lists (OpenAI/Anthropic/DeepSeek/Mistral +~8)	openai_provider.go:203, anthropic_provider.go:205, ...	SP1 (CONST-036)
D-6	Stub CLI-agent packages return hardcoded strings , never exec the binary	helix_agent/internal/clis/agents/qwencode/qwencode.go:101/115/137	SP4 (CONST-035)
D-7	helix_code/go.mod lacks replace dev.helix.agent → can't import the real clis/agent ic substrate	helix_code/go.mod	SP4/SP5
D-8	Constitution CLAUDE.md Revision-table drift	constitution/CLAUDE.md	SP0 resolved by sync (now Rev 21 @ f26368b)
D-9	Pooled dispatch methods are stubs too — return "<Agent> execution completed", no exec	helix_agent/internal/clis/instance_manager.go:906-942 (executeAider/ClaudeCode/Codex/Cline/OpenHands)	✅ FIXED (SP4, real os/exec, pin guards GREEN)
D-6	qwencode stub returns templated strings, never execs	helix_agent/.../qwencode/qwencode.go:101	✅ FIXED (SP4, real os/exec)
D-10	2 pre-existing flaky tests (nanosecond ID-collision)	helix_agent TestCompletionHandler_IDGeneration, TestGenerateTeamID	open (pre-existing, low-pri)
D-11	pre-existing "For now, allow all types" bluff	helix_agent/internal/clis/instance_manager.go:389 (IsAgentTypeAvailable)	open (SP4 follow-up)
D-12	~34 more execute* dispatch methods still stubbed	helix_agent/internal/clis/instance_manager.go (executeKiro... executeLlamaCode)	open (SP4 follow-up wave)
D-13	CONST-066 doc-export gap: 95+ docs/**/* .md (+ new specs/research) lack .html/.pdf siblings	repo-wide docs/**	open (D-13; F1 hook scoped too leniently — reconcile)

2. Sub-program decomposition

Each sub-program (SP) is independent: own spec, own plan, own evidence-backed release.
Dependency order shown.



SP0 — Constitution SYNC ONLY (DESCOPED per operator 2026-06-10)

Operator decision 2026-06-10: “We dont have need to change the constitution Submodule at all... just make constitution Submodule up to date, fetch and pull.” → **No §11.4.N authoring, no constitution push.** The reusable fork/merge scripts relocate to `scripts/caf/` **local to HelixCode** (not `constitution/scripts/`). Analysis-A’s drafted anchor is archived, unused (and §11.4.142 was in any case already taken upstream by the Universal-code-review mandate).

Status: DONE (this session). -  Fetched all 8 constitution remotes; **fast-forwarded** 60e2d66 → f26368b on main (0 ahead / 4 behind / ancestor → no loss, §11.4.113 no-force). - 
Recursive check: **no nested constitution submodules** (root-only, CONST-051 clean);
submodules/*/CONSTITUTION.md are cascade file-copies. -  New anchors now in force: **§11.4.142 Universal code-review** (every change → independent review before commit/build — applies to all SP implementation), **§11.4.143 video-journey** (N/A to HelixCode). - 
Pending follow-up: bump superproject submodule pointer 60e2d66 → f26368b (a meta-repo commit; itself subject to §11.4.142 review) — batch with next commit. -  **Requested but not yet run** (request “Before we begin”): CONST-055 post-pull full-rule validation sweep across project + submodules; correct any violation found. → tracked as **SP0.b**.

Original SP0 (superseded) — Constitution mandate + scripts

Goal: Land the new mandatory rule + generic reusable scripts in the constitution submodule BEFORE building on them (CONST-049 ordering). Feeds: analysis ... - A- (*Superseded by the*

descope above; the script designs from analysis-A/E are reused but land in scripts/caf/ local to HelixCode under SP3.)

- **P0.1 Fetch/pull constitution (deep, recursive).**
 - T0.1.1 `git -C constitution fetch --all --prune; pull; capture HEAD..@{u}.`
 - T0.1.2 Recurse: `git submodule foreach --recursive locate every nested constitution/governance copy; fetch/pull each.` Evidence: per-path log.
 - T0.1.3 Run `scripts/verify-all-constitution-rules.sh` (CONST-055) post-pull; record verdict.
- **P0.2 Audit project + submodules against ALL mandatory rules** (request §“Before we begin”).
 - T0.2.1 Run cascade verifier; enumerate any skipping/violation; open tracker items.
 - T0.2.2 Correct every detected violation (own sub-tasks per finding).
- **P0.3 Author the new fork-and-maintain anchor — §11.4.142** (next free number CONFIRMED by analysis-A; gate CM-COVENANT-114-142-PROPAGATION). ⚠ Constitution submodule is **detached HEAD @ 60e2d66** → `git checkout main` inside it FIRST; fix CLAUDE .md Revision-table drift (lags at Rev 23) per §11.4.44 in the same edit.
 - T0.3.1 Use analysis-A’s drafted anchor (forensic anchor = verbatim request lines 14–15; 6-clause body: fork+prefix → pointer-swap → recursive nested fork → automated upstream-merge → §11.4.28(B)-preserving edit allow-list [gitignore/remote-config/CI-disable/governance-pointers ONLY] → anti-bluff; composes §11.4.113 no-force-push).
 - T0.3.2 Cascade into `constitution/{Constitution,CLAUDE,AGENTS}.md + QWEN.md`; consumer-side index updates.
 - T0.3.3 Meta-test + paired §1.1 mutation for the propagation gate + 3 recommended per-family gates.
- **P0.4 Reusable generic scripts** in `constitution/scripts/` (decoupled CONST-051; inherited-by-reference §11.4.80 — never copied). Names per analysis-A:
 - T0.4.1 `fork_third_party_submodule.sh` (gh+glab fork-or-create under org+prefix caf-, params only, no force §11.4.113).
 - T0.4.2 `update_fork_from_upstream.sh` (fetch upstream main/master → merge into fork → push fast-forward-only).
 - T0.4.3 `resolve_recursive_fork_deps.sh` (recursive nested-dep handling; own-org → root per CONST-051; 3rd-party nested → fork).
 - T0.4.4 Anti-bluff Challenge + §1.1 mutation per script (throwaway repo bootstrap → assert remotes/merge).
- **P0.5 Commit + push constitution to ALL upstreams; bump pointer** (CONST-049 steps 4–7).

Done when: new anchor present + cascade-verified across fleet; scripts pass their Challenges with captured output; constitution pushed to all upstreams; meta-repo `.gitmodules` pointer bumped in same commit.

SP1 — Model-access refinement

Goal: For every provider, an API key supplied OR auto-recognized from `.env / .api_keys /` shell exports makes that provider’s **working** models available — dynamically obtained, validated, verified via LLMsVerifier. Feeds: analysis ... - B -

Verified facts (analysis-B): verifier consumption is REAL (verifier/client.go live HTTP; adapter.GetVerifiedModels:183 cache→live→stale→fallback). But: secrets.LoadAPIKeys (loader.go:30, reads api_keys.sh+.env) is **DEAD code** (zero production call-sites); no key-presence gate; the “working-model” predicate (Verified==true ∧ status=="verified" ∧ score≥GetMinAcceptableScore()) is loaded (adapter.go:175) but **never applied**; CLI handleListModels:1361 lists failed/pending as available. **The target flow already exists in helix_agent** (verifier/provider_types.go:349 SupportedProviders.EnvVars multi-alias + startup.go:389 per-key→DiscoverModels) + llms_verifier/api_keys/ → lift/extend per §11.4.74.

- **P1.1 Key auto-recognition.**

- T1.1.1 **Wire secrets.LoadAPIKeys at startup** in cmd/server + cmd/cli/main.go BEFORE config load (§11.4.124 — wire, don’t delete the dead code).
- T1.1.2 Multi-alias key-recognition table (reuse helix_agent SupportedProviders.EnvVars): .env → .api_keys/api_keys.sh → shell-exported precedence; per-provider detection.
- T1.1.3 Tests (unit: precedence + absence; integration: real key → provider enabled).

- **P1.2 Key-presence → provider-availability gate** — filter the verifier catalog by present keys (no key ⇒ provider hidden); fix factory.go:9 (single-provider) → enumerate every key-present provider.

- **P1.3 Dynamic working-model exposure via LLMsVerifier.**

- T1.3.1 Apply the unused “working-model” filter (key-present ∧ Verified==true ∧ status verified ∧ score≥min); poll ≤60s (CONST-038); verified ≤24h (CONST-037).
- T1.3.2 Replace hardcoded model lists (OpenAI:203, Anthropic:205, DeepSeek, Mistral, +~8) with verifier-sourced / dynamic fetch (CONST-036).
- T1.3.3 Define safe verifier-disabled/cold behaviour — FallbackModels (CLI:1387) must NOT be shown as “working” without validation (else §11.4 PASS-bluff).

- **P1.4 Docs/guides/manuals** for key setup + model availability + verification flow.

- **P1.5 Test coverage** (unit/integration/e2e/Challenge) — real keys where available, anti-bluff liveness (§11.4.107).

SP2 — HelixAgent exposure extension

Goal: Under one root expose: AI debate ensemble + HelixLLM (if enabled) + **every** discovered provider individually + each provider’s working models; update provider SDKs to latest. Naming fits existing conventions. Feeds: analysis ...-C-....

Verified facts (analysis-C): exposure surface = helix_agent/internal/router/router.go (Gin /v1): ensemble POST /v1/ensemble/completions:676, providers GET /v1/providers:773 (reads GetCapabilities().SupportedModels), discovery :1306, verification :1378; runtime registry services.ProviderRegistry (provider_registry.go:82). LLMProvider interface has **no** GetModels() — models only via GetCapabilities().SupportedModels. 48 hand-rolled HTTP providers, no vendor SDKs. HelixLLM buried as gated provider "helixllm" (USE_HELIX_LLM=true). “Working model” = DiscoveredModel.Verified == true (verifier). Building blocks exist; **no**

unified catalog joins ensemble + HelixLLM + each provider + each verified model. Inner Go module is at `helix_code/go.mod` (not `helix_code/helix_code/go.mod`).

- **P2.0 Bluff remediation:** `GET /v1/completion/models` (handlers/`completion.go:406`) returns a **hardcoded 3-model list** → replace with verifier-sourced (CONST-036 / BLUFF-002). RED→GREEN + §11.4.135 guard.
 - **P2.1 Unified catalog** joining ensemble + HelixLLM + each provider + each verified model as uniformly-named selectable targets (the missing layer).
 - **P2.2 Per-provider individual exposure** under the root alongside ensemble + HelixLLM.
 - T2.2.1 Provider catalog entry per discovered provider (verifier-sourced).
 - T2.2.2 Per-provider working-model list (`Verified==true`); consider adding `GetModels()` to the provider interface.
 - T2.2.3 Naming scheme (analysis-C): `ensemble, ensemble/confidence_weighted, helixllm, helixllm/helixllm-default, anthropic/claude-3-sonnet-20240229, openrouter/x-ai/grok-4` (reuses existing lowercase + vendor/model slash form).
 - T2.2.4 Promote HelixLLM to first-class root entry (not buried).
 - **P2.3 SDK/API currency** — deep web research (§11.4.99 latest-source) → update each provider SDK to latest (bedrockruntime/azcore/azidentity/tree-sitter pin; resolve gin 1.11.0[?] 1.12.0 skew across `helix_code/helix_agent`); bridged CLI agents + their models up to date.
 - **P2.4 Docs/guides/manuals/graphs/schemes/SQL** + website + video-course updates (request-mandated deliverables).
 - **P2.5 Full test coverage.**
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SP3 — Fork-ALL mechanism + auto-merge

Goal: Fork every `cli_agents/*` under `vasic-digital/caf-*`, swap submodules to forks, recursively handle nested deps, auto-merge upstream regularly. Feeds: analysis ... - E - Uses SP0 scripts.

Verified facts (analysis-E, parsed from 381-line .gitmodules): 50 fork entries (49 top-level + 1 nested). **EXCLUSIONS:** `cli_agents/claude-code-source` is already the operator's own GitLab mirror (milos85vasic/ccode-private) → **skip fork** (default-excluded). Only ONE nested submodule fleet-wide: `cline/.gitmodules` → `cline-bench` (HTTPS → SSH `caf-cline-bench`); max depth 1, no own-org nested chains (CONST-051 clean). OpenAI-Cookbook uses `org-NNNN@github.com`: alias (out of scope, resources dir) — parser must normalize.

- **P3.1 Fork mapping table** — built (49 GitHub upstreams → `vasic-digital/caf-<name>`, kebab names preserved); GitHub+GitLab dual-remote.
- **P3.2 Execute fork-all** via the SP0 fork script (operator-gated; **high blast radius** → §11.4.101 **explicit confirm before any external mutation**; async fork-readiness polling).
- **P3.3 Submodule swap** (`.gitmodules` URL rewrite + `git submodule sync --recursive`), per-fork validate buildable.
- **P3.4 Nested-submodule resolution** (just `cline-bench` today; own-org → root per CONST-051; 3rd-party nested → fork).
- **P3.5 Auto-merge automation** (scheduled merge-into-fork-only, **no-force** §11.4.113, clean commits §11.4.30).
- **P3.6 Build + install each locally** with correct git-ignore hygiene (no stray files committed).

- **P3.7 Reconcile script naming** between analysis-A (`fork_third_party_submodule.sh`) and analysis-E (`caf_fork_all.sh + caf_lib.sh`) into one unified, decoupled set before authoring.
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SP4 — CLI-agent bridge providers (one provider per agent)

Goal: Each working CLI agent = its own provider proxying prompts via CLI + exposing power-features (Vision/Generative/RAG/Memory/MCP/LSP/ACP). Primary = system-installed; fallback = submodule-built. Re-export+install+validate every CLI config. Feeds: analysis ... - D -

Verified facts (analysis-D): 8 agents system-installed NOW (claude/qwen/opencode/gemini/crush/codex/goose/copilot — versions captured). The real `exec.LookPath` orchestration lives in `helix_agent/internal/clis` (`agents/base/base.go:120`, `instance_manager.go`), NOT in `helix_code/internal/llm` (its `copilot_provider.go` is HTTP/token, not CLI-exec). `llm.Provider` contract at `missing_types.go:356`. Re-export exists (`helixagent --generate-all-agents,main.go:4522`).

- **P4.0 Pre-req remediation (anti-bluff, BLOCKS the rest).**
 - T4.0.1 **Fix stub per-agent packages** in `helix_agent/internal/clis` that return hardcoded strings without `exec` (e.g. `agents/qwencode/qwencode.go:101/115/137`) — these are live §11.4 / CONST-035 bluffs. RED-on-stub → GREEN-on-real-exec (§11.4.115).
 - T4.0.2 Add `replace dev.helix.agent` wiring so `HelixCode` can import the `clis` layer (or thin adapter per CONST-051/§11.4.74).
 - **P4.1 Per-agent provider contract** — one parametrized `CLIAgentProvider` implementing `llm.Provider`; reuse the `clis` layer via adapter, don't reimplement.
 - **P4.2 Primary-vs-fallback selection** (real `exec.LookPath` system-installed → submodule-built).
 - **P4.3 Power-feature passthrough** (RAG/Memory/MCP/LSP/ACP/Vision) + dynamic `GetModels` per agent (CONST-036) + capabilities from `LLMsVerifier` (CONST-040) — research per agent (§11.4.99): `OpenCode/Zen` non-interactive flags, `Claude Code -p/- -output-format`, `Qwen Code`.
 - **P4.4 Config re-export+install+validate** — re-export exists; **add the ABSENT filesystem-install + LIVE post-install validation** (each installed config drives a real proxied prompt → real result, captured evidence). Note: current configs point agents *at* `HelixAgent` (reverse) — D adds the forward bridge direction.
 - **P4.5 Tier-1 agents first** (the 8 installed) → expand to ~43 candidates.
 - **P4.6 Full test coverage** per agent (real proxied prompt → real result, anti-bluff §11.4.98 fully automated).
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SP5 — Parallel / subagent-driven enforcement

Goal: All `HelixCode` work can be done by coordinated parallel agents; subagents-driven used maximally without harming the main stream. Feeds: analysis ... - F - As a new `HelixDevelopment` submodule if warranted.

Verified facts (analysis-F): capability exists twice, duplicated not shared — `HelixCode(internal/agent/coordinator.go:11-50 registry+queue+circuit-breakers+voting, agent/subagent/in-process+subprocess+worktree spawners, worker/SSH pool, workflow/DAG)`

vs HelixAgent(internal/agent/workflow.go:28 LangGraph-style, agents/swarm/, ensemble/, planning/{tree_of_thoughts,mcts,hiplan}). **GAP:** helix_code/go.mod doesn't import the own-org agentic/concurrency/planning substrate; no universal "every operation is a parallel-dispatchable agent unit."

- **P5.1 Map existing multi-agent/worker/queue capability.** (done — analysis-F)
 - **P5.2 De-duplicate into a shared substrate** (reuse before reimplement §11.4.74) — universal parallel-dispatchable agent-unit abstraction.
 - **P5.3 Coordination layer** (dispatch, isolation, quiescence §11.4.84, PWU §11.4.58, §11.4.103 ≥ 3 streams).
 - **P5.4 Wire into HelixCode + HelixAgent** (add the missing own-org imports).
 - **P5.5 Full test coverage** (stress/chaos §11.4.85 for concurrency).
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SP6 — Other execution models (dynamic flows, ...)

Goal: Dynamic flows + other execution models as NEW decoupled public submodules under HelixDevelopment, wired into HelixCode + HelixAgent. Feeds: analysis ... - F -

Verified facts (analysis-F): existing models = DAG workflow (HelixCode), graph/state-machine (helix_agent/agent), ToT/MCTS/hierarchical planning.
ABSENT as named packages: pipeline, standalone dag, dataflow, reactive, saga, behavior-tree. Reuse candidates already in tree. Proposed NEW HelixDevelopment submodules: **flow_engine**, **dag_orchestrator**, **pipeline_runtime**, **agent_mesh/swarm_kit**, **flow_dsl**; 12 deep-research topics enumerated.

- **P6.1 Deep web research** on execution models (dynamic flow + others) — 12 topics in analysis-F (§11.4.99 latest-source).
 - **P6.2 Create public submodules** (gh+glab) under HelixDevelopment: flow_engine / dag_orchestrator / pipeline_runtime / agent_mesh / flow_dsl; decoupled (CONST-051), helix-deps.yaml (CONST-054). Reuse existing own-org pieces first (§11.4.74).
 - **P6.3 Integration analysis + wiring** into both consumers (shared Node/Scheduler/Dispatcher/Resolver interfaces + go.mod replace).
 - **P6.4 Full test coverage + Challenges.**
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SP7 — Testing & QA (cross-cutting)

Goal: 100% coverage across every supported test type + Challenges + HelixQA banks + fully-autonomous QA sessions; no bluff; independent-agent confirmation (CONST-050/048/098). Feeds: analysis ... - F -

Verified facts (analysis-F): HelixCode strong on unit/integration/e2e/security/chaos/stress/performance/benchmarking + §1.1 mutation meta-tests (tests/stresschaos/stresschaos_meta_test.go) + regression dir. **HelixCode-local GAPS:** ddos, scaling, ux have no local harness (HelixQA/challenge-only); ui thin → delegation risks a coverage bluff unless evidence-backed. **⚠ HelixQA is STALE** (README round 219 / 2026-05-19).

- **P7.1 Update HelixQA** submodule to latest (fetch/pull + pointer bump).

- **P7.2 Test-type coverage matrix** (unit/integration/e2e/full-automation/security/ddos/scaling/chaos/stress/performance/benchmark/ui/ux/Challenges/HelixQA) × every new feature.
- **P7.3 Autonomous QA sessions** executing every registered bank with captured wire evidence.
- **P7.4 Independent code-review agents** confirm each closure (§11.4.125/§11.4.134).
- **P7.5 Regression-guard registration** for every fixed defect (§11.4.135).

3. Cross-cutting mandatory constraints (apply to every SP)

- **Anti-bluff §11.4 family** — captured evidence on every closure; no metadata-only PASS.
- **No-force-push §11.4.113**; merge-onto-latest-main for every repo/submodule.
- **CONST-049** ordering for all constitution edits.
- **Subagent-driven default §11.4.70 / §11.4.103** — ≥ 3 parallel background streams while actionable items exist.
- **Fully-automated tests §11.4.98; rock-solid-proof-or-research §11.4.123.**
- **§11.4.122** — no silent removal of existing capabilities without operator confirmation.
- **CONST-046** — no hardcoded user-facing content; **CONST-036** — no hardcoded model lists.
- **Continuation §13.1 (CONST-044)** kept in sync each state-advancing commit.

4. Detailed implementation plans (written 2026-06-10)

SP	Plan doc	Scope
SP1	plans/2026-06-10-SP1-model-access-plan.md	6 phases / 18 TDD tasks / 4 decisions
SP2	plans/2026-06-10-SP2-helixagent-exposure-plan.md	7 phases / 21 tasks / 6 decisions
SP3	plans/2026-06-10-SP3-fork-mechanism-plan.md	50-fork map + 3 scripts at scripts/caf / tasks T3.0–T3.7 / G-1 gated
SP4	plans/2026-06-10-SP4-cli-bridge-plan.md	7 phases / 34 tasks / 4 decisions
SP5+SP6	plans/2026-06-10-SP5-SP6-parallel-dynamic-plan.md	reuse-first; create 2 repos / 5 decisions
SP7	plans/2026-06-10-SP7-testing-qa-plan.md	phases A–E / 6 gaps / 5 decisions

5. Consolidated decision register

(a) **Operator-GATED** — irreversible / high-blast / genuinely ambiguous (I will NOT proceed without explicit go): - **G-1** Execute **fork-ALL 49 repos** under vasic-digital/caf - * (SP3) — irreversible external.  **PRE-AUTHORIZED** by operator 2026-06-10 — may run once the SP3 scripts pass their anti-bluff Challenge (no re-ask). - **G-2** Create **2 new HelixDevelopment repos** dag_orchestrator + pipeline_runtime (SP6) — irreversible external. **Still requires explicit go.** - **G-3** **SDK-update blast radius** (SP2) —

bumping bedrockruntime/azcore/tree-sitter + resolving gin 1.11.0[?]1.12.0 across both modules can break builds → operator scope/sequence call. - **G-4 Video-course / website production scope** (SP2 OP-6) — defer vs in-scope. - **G-5** Whether to **de-bluff all ~70 agent packages now** vs tier-1-first + track-rest (SP4 D-C; D-6/D-9 live in `helix_agent`, a CONST-047 submodule).

(b) Reversible / clear-default — I'll proceed on the subagents' recommended defaults unless you object (§11.4.101): - Lift a decoupled key-recognition table from `helix_agent` `SupportedProviders.EnvVars` (SP1); `LoadAPIKeys` precedence = gap-fill (don't override already-exported shell env). - `Verifier-disabled` = honest empty + opt-in - `include-unverified` live path (never show `FallbackModels` as “working”). - Unified catalog at `internal/catalog/` + `GET /v1/catalog`; promote `HelixLLM` to first-class; naming `ensemble//helixllm//<provider>/<model>`. - De-harden funnel-first, then `OpenAI/Anthropic/DeepSeek/Mistral`. - SP5 substrate = `EXTEND Concurrency/Agentic, REUSE Planning, FOLD agent_mesh` in; first task adds the `replace` directives (closes D-7). **[M-1 fix] SP5 T5.2 is the SOLE owner of the replace dev.helix.agent + replace dev.helix.substrate helix_code/go.mod edits; SP4 T4.0.x CONSUMES them as a prerequisite (depends-on SP5 wiring) and NEVER re-edits go.mod** — resolves the parallel-stream edit-collision. - New local harnesses for `ddos/scaling/ux/ui` with anti-delegation-bluff gates; `HelixQA` fetch-to-latest (§11.4.71 merge-onto-main).

5a. Bundle code-review (§11.4.142) — verdict + fixes applied

Independent review 2026-06-10-planning-bundle-review.md: **GO-WITH-FIXES** (0 BLOCKER · 1 MAJOR · 5 MINOR); 18 anchors spot-checked incl. all of D-1..D-9 — zero fabricated/drifted. Fixes: - **M-1 (MAJOR) — FIXED:** `replace dev.helix.agent/ dev.helix.substrate go.mod` edits sole-owned by SP5 T5.2; SP4 consumes as prerequisite (see §5(b)). - **m-1 — N/A:** analysis-A's §11.4.142 choice is moot (SP0 constitution authoring descoped; upstream already has §11.4.142). - **m-2 — noted:** SP4's “beyond roadmap” label for D-9 is cosmetic; D-9 is in the ledger (§1a). - **m-3 — TODO in SP7:** add a D-9 regression-guard row to the registry (carried into SP7 execution). - **m-4 — RESOLVED:** SP3 plan now written. - **m-5 — tracked:** CONST-055 full-rule audit = SP0.b (still unrun; offered to operator).

6. Next actions

1.  Planning complete: 6 analyses + 5 SP plans + defect ledger D-1..D-9. SP3 plan + planning-bundle code-review in flight.
2.  **HERE:** operator confirms the **decision register** (esp. the 5 gated items G-1..G-5).
3. Commit the planning bundle (needs §11.4.142 independent review GO + §11.4.75 `.html/.pdf` siblings + superproject pointer bump) — **operator: commit now?**
4. On go: implement RED-first per SP, each change through the §11.4.142 review gate; parallel non-contending streams (§11.4.103); irreversible externals only after G-1/G-2 go.